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Exercise 1

Variables, Collections & Conditionals

Data Analytics with Python · Databricks Environment

```
student_name age subject score passed
# "Amara" 19 "Math" 72 True
# "Ben" 21 "Science" 55 False
# "Chloe" 20 "Math" 88 True
# "David" 22 "English" 45 False
# "Eva" 19 "Science" 91 True
# "Farai" 23 "English" 63 True
# "Grace" 20 "Math" 50 False
# "Hassan" 21 "Science" 77 True
```

```
student_name="Amara"
age=19
subject="math"
score=72
passed=True
```

```
student_name:str
age:int
score:int
passed:bool
<class 'int'> <class 'bool'>
```

```
#What is Amaras final mark?
score =72
```

```
final mark = (score / 100) * 150
print (final mark)
108.0
```

#Create a list called student names that contains all 8 student names from the dataset, in order.

```
student names = ["Amara", "Ben", "Chloe", "David", "Eva", "Farai", "Grace", "Hassan"]
student names [0] = "Amara"
student names [-1] = "Hassan"
student names [2:5] = ['Chloe', 'David', 'Eva']
Len (student names) == 8
```

True

```
score= [71,55,88,45,91,63,50,77]
score [0] =72
score [4] =91
score [0] +score [4]
```

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#Create a dictionary called hassan that stores all five fields for Hassan as key-value pairs.

```
hassen= {"student_name": "Hassen", "age": 21, "subject": "sciens", "score": 77, "passed": True}
```

```
hassen["subject"] == "science"
hassen["score"] + 10 == 87
type(hassen["passed"]) == bool
```

True

```
students = [
    {"student_name": "Amara", "age": 19, "subject": "Math", "score": 72, "passed": True},
    {"student_name": "Ben", "age": 21, "subject": "Science", "score": 55, "passed": False},
    {"student_name": "Chloe", "age": 20, "subject": "Math", "score": 88, "passed": True},
    {"student_name": "David", "age": 22, "subject": "English", "score": 45, "passed": False}
]
```

```
students [1] ["score"] ==55
```

```
students[0]["student_name"] == "Amara"  
print(len(students))
```

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```
score = 72  
age = 19  
subject = "Math"  
passed = True
```

```
score > 60 -- True  
age == 20 -- False  
subject != "science" # True  
score >= 72 -- True
```

```
score > 60 and passed == True # True  
age < 18 or score > 70 # True  
not passed # False  
score > 80 and subject == "Math" # False
```

False

```
# Amara's score  
score = 72  
name = "Amara"  
  
if score >= 75:  
    print(name + ": Distinction")  
elif score >= 50:  
    print(name + ": Pass")  
else:  
    print(name + ": Fail")
```

```
# David's score  
score = 45  
name = "David"  
  
if score >= 75:  
    print(name + ": Distinction")  
elif score >= 50:  
    print(name + ": Pass")  
else:  
    print(name + ": Fail")
```

Amara: Pass David: Fail

#Write your if/elif/else code below and record the output for each test.

```
score = 72
if score >= 75:
    print("Distinction")
elif score >= 50:
    print("Pass")
else:
    print("Fail")
```

Pass

#output when score=72-passed

#output when score=45-final

#output when score=91-distinction

```
score = 72
subject = "Math"

if score > 60 and subject == "Math":
    print ("Math high achiever")
else:
    print ("Does not meet criteria")
```

Math high achiever

#Test with Amara (score=72, subject="Math") and Ben (score=55, subject="Science")

```

# Amara
score = 72
name = "Amara"
subject = "Math"

if score >= 75:
    print (name + " (" + subject + "): Distinction")
elif score >= 50:
    print (name + " (" + subject + "): Pass")
else:
    print (name + " (" + subject + "): Fail")

```

```

# Ben
score = 55
name = "Ben"
subject = "Science"

if score >= 75:
    print (name + " (" + subject + "): Distinction")
elif score >= 50:
    print (name + " (" + subject + "): Pass")
else:
    print (name + " (" + subject + "): Fail")

```

Amara (Math): Pass Ben (Science): Pass

#The logical operator is used in the conditional statement is the greater than or equal to (>=)

#R1
#Alist stores multiple items in order, while a dictionary stores data as key-value pairs.

#Example of List
scores = [72, 55, 88]

#Example of dictionary
student = {"name": "Amara", "score": 72}
#I would use a list when storing many values in order.
I would use a dictionary when I need labels or keys for the data.

#R2

Explain in plain English what this expression checks: `score > 60 and passed == True`

`score > 60` and `passed == True`

#check whether the student scored more than 60 and also passed the exam.

True

#R3

If you wanted to store all 8 students and be able to look up any student by name quickly, which collection type would you use? Why?

#I would use a dictionary because dictionaries allow quick lookup using a student's name as the key.

#for example.

```
students = {"Amara": {"score": 72},  
           "Ben": {"score": 55}}
```